From measurements to activities and fitted models

Read and prepare: 15–20 minutes total. Bring tentative answers; uncertainty is useful for discussion.

What evidence makes an activity-model fit useful?

Activity has a reference

Chemical potential can be written $\mu_i = \mu_i^\circ + RT \ln a_i$. Activity is dimensionless and carries a chosen standard state. For a liquid using the pure-liquid standard at the same T and P, $a_i=\gamma_i x_i$. The coefficient γ describes departure from the ideal-solution reference; it is not the activity itself. Comparing coefficients requires the same component order and convention.

Residual and excess are different comparisons

A residual property compares a real fluid with the corresponding ideal gas at the same T, P and composition. An excess property compares a real mixture with an ideal solution at the same T, P and composition. EOS calculations and activity models therefore use different reference comparisons. For the chosen liquid convention, $g^E/(RT)=\Sigma x_i \ln \gamma_i$. Choosing NRTL or Margules is choosing a description of this nonideality.

Predict first, infer second

With a model and its parameters, predict pressure and vapor composition. With measured T, P, x and y, invert the ideal-vapor relation to infer activities, then fit model parameters. First check the vapor-pressure equation, logarithm base, units and temperature range. A wrong Psat can be absorbed into fitted parameters and leave a deceptively small residual.

$\gamma_i = y_i P / [x_i P_i^{sat}(T)]$ (ideal vapor, negligible Poynting correction)

$a_i = \gamma_i x_i$; $g^E/(RT) = \Sigma x_i \ln \gamma_i$

One example to reason through

Synthetic teaching values. These numbers illustrate the assumptions; they are not measurements of a real material.

At a synthetic measured point P=100 kPa, x1=0.4 and y1=0.6, choosing Psat1=120 kPa gives γ1=1.25 and a1=0.50. Substituting Psat1=150 kPa instead gives γ1=1.00 and a1=0.40. The mixture measurements have not changed. This calculation isolates the effect of the pure-component property input. It does not show which Psat is physically correct; that requires an applicable source.

Keep three questions separate

Convergence: did the numerical algorithm satisfy its stopping criterion?

Consistency: are the observations compatible with thermodynamic constraints under the selected assumptions?

Prediction: how well does the model describe observations excluded from fitting?

Check and interpret

Numerical convergence, thermodynamic consistency and predictive accuracy answer different questions. Inspect residual patterns and model assumptions. An area diagnostic depends on coverage and endpoint treatment. Held-out error uses data excluded from fitting. Neither a low fitting error nor one consistency test certifies the dataset.

Bring three short answers

A sketch, a few numbers or one sentence is enough. Do this before opening the calculator; revise your prediction in class.

1. Retrieve Week 1: at 350 K and z1=0.5 with Psat1=150 and Psat2=60 kPa, predict the phases at 80, 95 and 110 kPa. At 95 kPa check $z1=(1-0.493506)(0.388889)+0.493506(0.614035)$.

2. For P=100 kPa, x1=0.4, y1=0.6 and Psat1=120 kPa, show the substitution for γ1 and a1. Explain why activity differs from its coefficient.

3. List the Antoine form, log base, temperature and pressure units and valid range. Could a flexible fit conceal an incorrect Psat?

Optional extension

Choose a controlled consistency defect in Lab 03, a fixed held-out split, or conditional bootstrap spread in Lab 08. Fixed alpha and property assumptions remain outside that bootstrap. Exact synthetic data can produce almost zero spread. No real-data upload is required before class. Use week02-practice.pdf for the English checklist and independent-study record; it is the same task, not additional homework.

Continue in the core lab

https://www.skhgroup.net/teaching/2105603/labs/vle-input.html

Reading locator: supplied textbook contents, sections 11.4, 11.8-11.11, 14.1-14.4, 14.8, 15.14-15.18.

Original course explanation and examples, not a reproduction of the textbook. Section numbers are based on the supplied contents pages. Required weekly independent work: 15–20 minutes preparation plus 30–40 minutes follow-up; optional extensions are additional.